

*Patuakhali Science and Technology University*  
*Faculty of Computer Science and Engineering*

## **Project Proposal on** **KajBazar: A community-Driven Service Provider** **Directory Platform**

**Course Title:** System Analysis and Design Sessional  
**Course Code:** CIT - 222

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# **KajBazar: A Community Driven Service Provider Directory Platform**

## **1. Problem Statement**

Local skilled workers such as electricians, plumbers, carpenters, mechanics, and other service providers mainly rely on word-of-mouth to find customers. As a result, many qualified workers remain difficult to discover, while consumers often struggle to find reliable service providers in their area. The lack of a centralized and trusted platform also makes it difficult to verify workers' credibility or compare their experience and customer feedback.

KajBazar is proposed as a web-based(for now) Community Service Provider Directory that connects consumers with local skilled workers. The platform enables users to search providers by category and location, view verified profiles, contact workers directly, and share their experiences through ratings and reviews. It also allows community members to recommend offline workers, helping expand the directory and increase employment opportunities for local service providers.

## **2. Objectives**

### **2.1 General Objective**

To design and develop KajBazar, a web-based community service provider directory that connects consumers with verified local skilled workers and enables users to easily search, contact, and review service providers.

### **2.2 Specific Objectives**

- To provide a categorized directory of local skilled workers such as electricians, plumbers, carpenters, mechanics, and other service providers.
- To enable consumers to search workers based on service category and location.

- To allow workers to create and manage verified profiles with their skills, experience, and contact information.
- To provide a direct contact system that allows consumers to communicate with workers without intermediaries.
- To implement a rating and review system that helps consumers evaluate service quality.
- To allow community members to recommend skilled offline workers for inclusion in the platform.
- To provide an administrator dashboard for managing users, worker verification, categories, reviews, and reported content.

### 3. Feasibility Study

| Dimension          | Assessment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Technical</b>   | The proposed system will be developed using <b>HTML, CSS, JavaScript, React.js</b> for the frontend, <b>ASP.NET Core with Entity Framework Core</b> for the backend, <b>PostgreSQL</b> as the database management system, and <b>JWT Authentication</b> for secure user authentication and authorization. These technologies are modern, well-documented, widely adopted in the industry, and provide the necessary scalability, security, and maintainability to successfully implement the system within the project timeline. |
| <b>Operational</b> | KajBazar digitizes the traditional process of finding local skilled workers by providing a centralized and user-friendly platform. Consumers can easily search for workers based on service category and location, while service providers can create and manage their profiles efficiently. The simple interface and direct-contact approach ensure that consumers, workers, and administrators can use the system with minimal training or technical knowledge.                                                                |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Economic</b> | The system utilizes open-source frameworks, libraries, and development tools, significantly reducing implementation costs. Since no specialized hardware or expensive software licenses are required, the primary operational expenses are web hosting and database services, which are affordable and suitable for small- to medium-scale deployment. This makes the project economically feasible for both academic and practical implementation.            |
| <b>Schedule</b> | The initial version of KajBazar focuses on core functionalities, including worker registration, profile management, service search, location-based filtering, ratings and reviews, direct contact, and administrative management. By limiting the scope to these essential features, the project can be realistically designed, developed, tested, and documented within the allocated academic semester while allowing future enhancements in later versions. |

## 4. Information Gathering

Requirements for KajBazar will be collected through a combination of qualitative and user-centered information gathering techniques to ensure that the system effectively addresses the needs of both consumers and local service providers. These techniques will help identify existing problems, understand user expectations, and validate the proposed features before development.

**Interviews:** Structured interviews will be conducted with local skilled workers such as electricians, plumbers, carpenters, mechanics, painters, and other service providers, as well as potential consumers. The interviews will focus on understanding the difficulties workers face in reaching customers, the challenges consumers experience when searching for reliable service providers, and the features they expect from a digital platform.

**Observation:** Existing methods of finding and hiring local workers, including word-of-mouth recommendations, neighborhood referrals, social media groups, online marketplaces, and local advertisements, will be observed to identify their strengths, limitations, and common inefficiencies. This analysis will help determine how KajBazar can provide a more organized and accessible solution.

**Discussion with Service Providers:** Focus group discussions will be conducted with service providers to gather detailed requirements regarding worker registration, profile management, identity verification, service categorization, availability, and customer communication. Their feedback will also help ensure that the platform remains simple and beneficial for workers with varying levels of technical experience.

**Consumer Feedback:** A structured questionnaire will be distributed among potential users from different communities to understand their expectations regarding worker search, location-based filtering, profile information, direct contact, ratings and reviews, and overall platform usability. The collected feedback will be analyzed to prioritize features and improve the user experience in the final system design.

## 5. Benchmark Analysis

KajBazar shares similarities with existing online service platforms by allowing users to discover and contact skilled service providers. However, unlike general service marketplaces, KajBazar focuses on connecting consumers directly with verified local workers through a simple directory-based approach. The table below compares KajBazar with three widely used service platforms.

| Feature                          | KajBazar     | Sheba xyz | Handy | Thumbtack |
|----------------------------------|--------------|-----------|-------|-----------|
| Worker search by category        | Yes          | Yes       | Yes   | Yes       |
| Location-based search            | Yes          | Limited   | Yes   | Yes       |
| Verified worker profiles         | Yes          | Yes       | Yes   | Yes       |
| Direct contact with workers      | Yes (Unique) | No        | No    | Limited   |
| Community worker recommendations | Yes (Unique) | No        | No    | No        |

|                                         |     |         |     |     |
|-----------------------------------------|-----|---------|-----|-----|
| Ratings & Reviews                       | Yes | Yes     | Yes | Yes |
| In-app Messaging                        | Yes | Yes     | No  | Yes |
| Service request posting                 | Yes | Yes     | No  | Yes |
| Administrator verification & management | Yes | Limited | Yes | Yes |

Although Sheba xyz, Handy, and Thumbtack provide platforms for finding service providers, they primarily operate as service marketplaces where requests are managed through the platform. KajBazar differs by functioning as a community-driven service provider directory that allows consumers to directly contact verified workers without intermediaries. Additionally, its **Community Worker Recommendation** feature enables users to recommend skilled offline workers for inclusion in the directory, helping expand employment opportunities while maintaining a trusted and locally focused service network.

## 6. Feature List

### 6.1 User Registration & Authentication

KajBazar provides secure registration and login facilities for consumers, service providers, and administrators. JWT-based authentication and role-based access control ensure secure access to the system.

### 6.2 Worker Directory & Search

Consumers can browse and search verified service providers by category and location. The system also supports filtering workers based on ratings and experience to help users find suitable professionals.

### 6.3 Worker Profile Management

Service providers can create and maintain professional profiles containing their skills, experience, service areas, contact information, and availability. Administrators verify profiles before they become publicly visible.

## 6.4 Direct Contact & Reviews

Consumers can directly contact workers using the provided contact information without any intermediary. After receiving services, users can submit ratings and reviews to improve transparency and help others make informed decisions.

## 6.5 Community Recommendation

Users can recommend skilled offline workers who are not yet registered on the platform. Administrators review these recommendations before adding new workers to the directory.

## 6.6 Administrative Management

Administrators can manage users, worker profiles, service categories, reviews, reports, and community recommendations through a centralized dashboard, ensuring the platform remains secure, organized, and reliable.

# 7. System Requirements

## 7.1 Hardware Requirements

| Component     | Minimum Specification                                                                      |
|---------------|--------------------------------------------------------------------------------------------|
| Server        | Quad-core CPU, 8 GB RAM, 100 GB SSD Storage                                                |
| Client Device | Desktop, Laptop, Tablet, or Smartphone with a modern web browser and internet connectivity |
| Network       | Stable broadband internet connection (minimum 5 Mbps recommended)                          |

## 7.2 Software Requirements

| Layer                     | Technology                                  |
|---------------------------|---------------------------------------------|
| Frontend                  | HTML5, CSS3, JavaScript, React.js           |
| Backend                   | ASP.NET Core (.NET 8)                       |
| Database                  | PostgreSQL                                  |
| ORM                       | Entity Framework Core                       |
| Authentication            | JSON Web Token (JWT)                        |
| Operating System (Server) | Windows Server or Linux with .NET 8 Runtime |
| Development Tools         | Visual Studio 2022 / Visual Studio Code     |
| Version Control           | Git & GitHub                                |
| API Testing               | Postman                                     |

## 8. Business Rules

**BR-01.** Every user must register and authenticate before accessing role-specific features of the system.

**BR-02.** A service provider must complete profile registration before applying for verification.

**BR-03.** Only verified service providers are displayed in the public worker directory.

**BR-04.** A worker can belong to one or more service categories but must provide valid information for each profile.

**BR-05.** Consumers can search and filter workers based on service category, district, and sub-district.



**BR-06.** Consumers can directly contact verified workers using the contact information available in their profiles.

**BR-07.** Only registered consumers who have contacted a worker can submit ratings and reviews.

**BR-08.** A consumer may submit only one review for the same worker unless the previous review is updated.

**BR-09.** Users may recommend skilled offline workers, but only administrators can approve and add them to the directory.

**BR-10.** Only administrators can verify, suspend, or remove worker profiles from the platform.

**BR-11.** Administrators are responsible for managing service categories, districts, and sub-district information.

**BR-12.** Reviews or user reports that violate platform policies may be edited or removed only by an administrator.

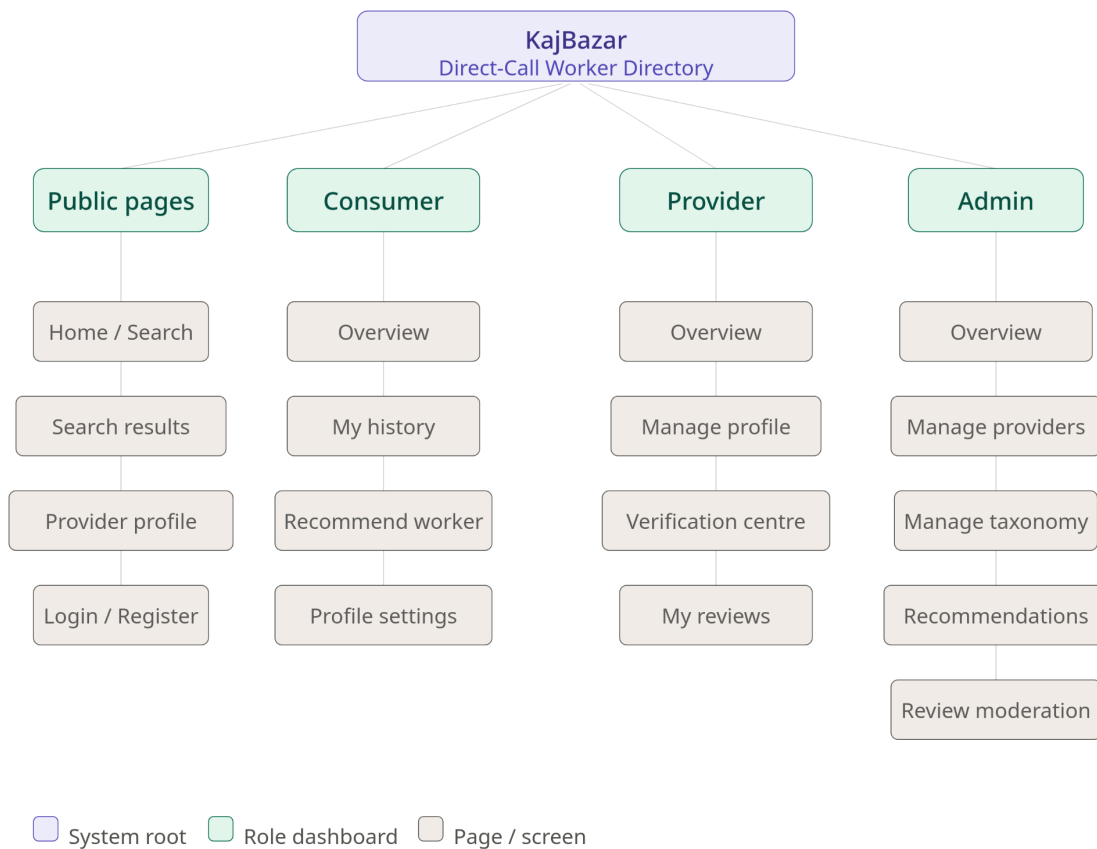
**BR-13.** All user accounts must be authenticated before accessing any protected dashboard or performing restricted operations.

**BR-14.** The system must maintain records of worker registrations, profile updates, reviews, and administrative actions for future reference and auditing.

**BR-15.** Personal information and user data must be securely stored, and access shall be restricted according to user roles and system permissions.

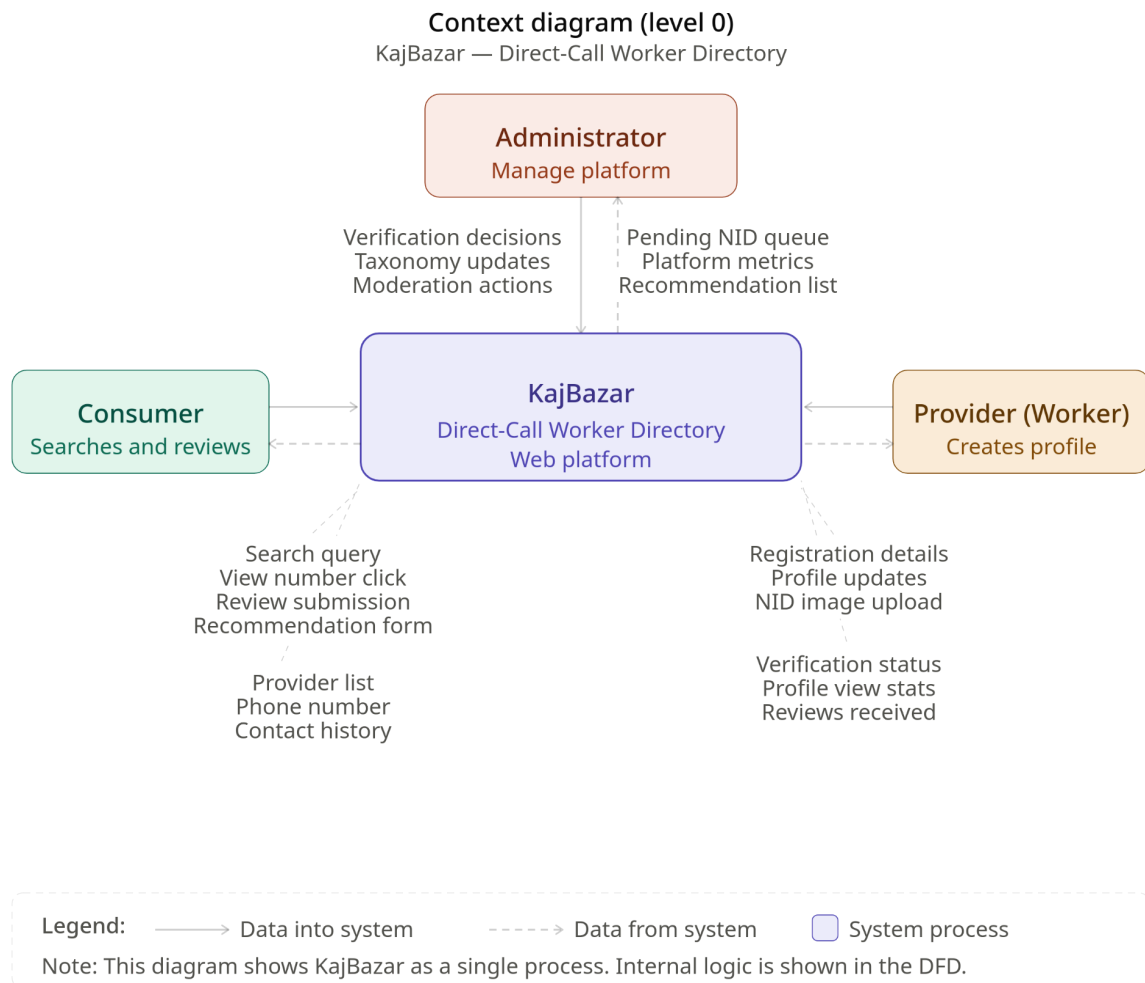
## 9. Site Map

The site map below illustrates the hierarchical structure of KajBazar, showing the main pages and navigation flow for Consumers, Service Providers, and Administrators.



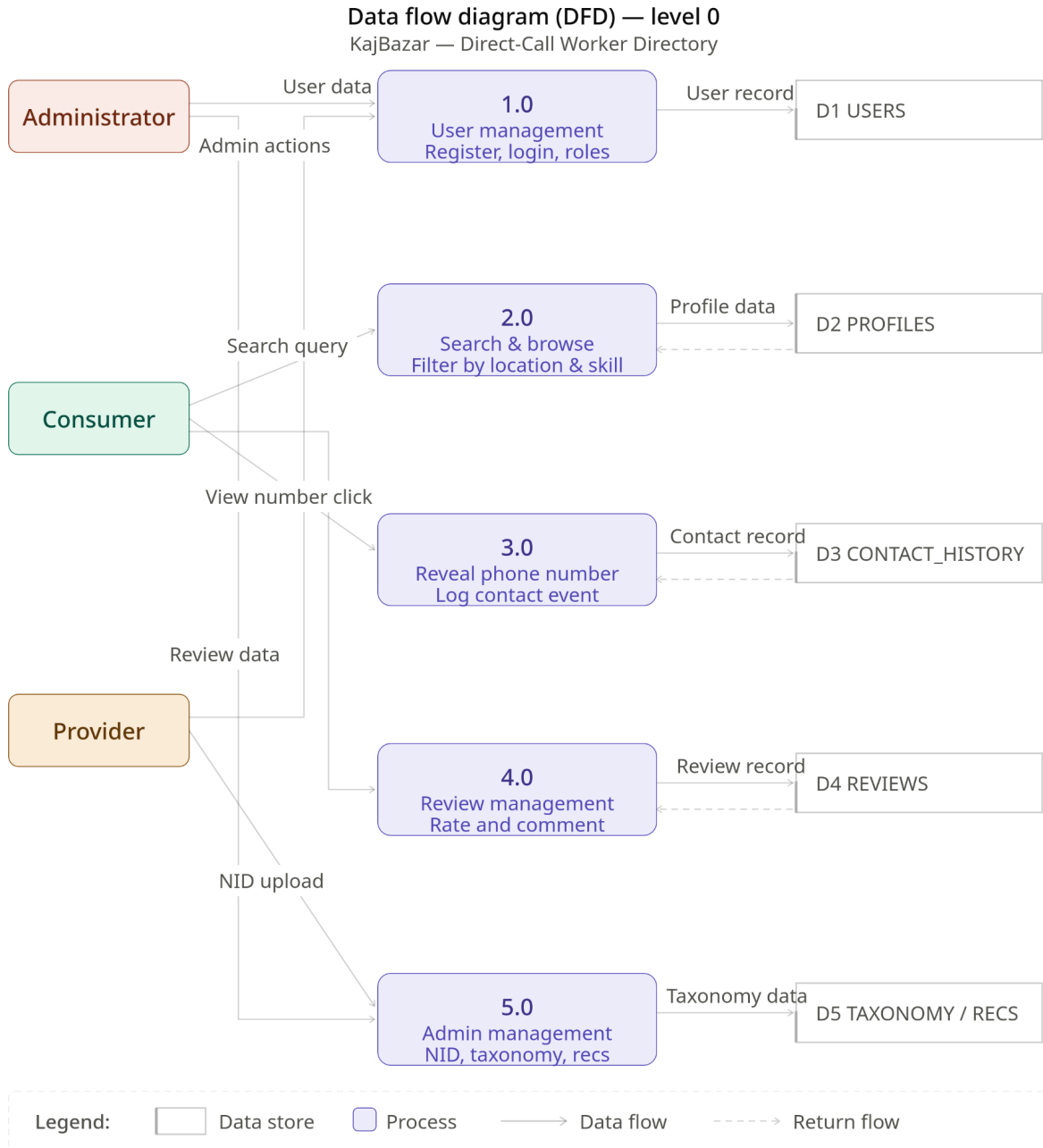
## 10. Context Diagram (Level 0)

The context diagram illustrates KajBazar as a single process that exchanges information with its three external entities: Consumer, Service Provider, and Administrator.



# 11. Data Flow Diagrams (DFD)

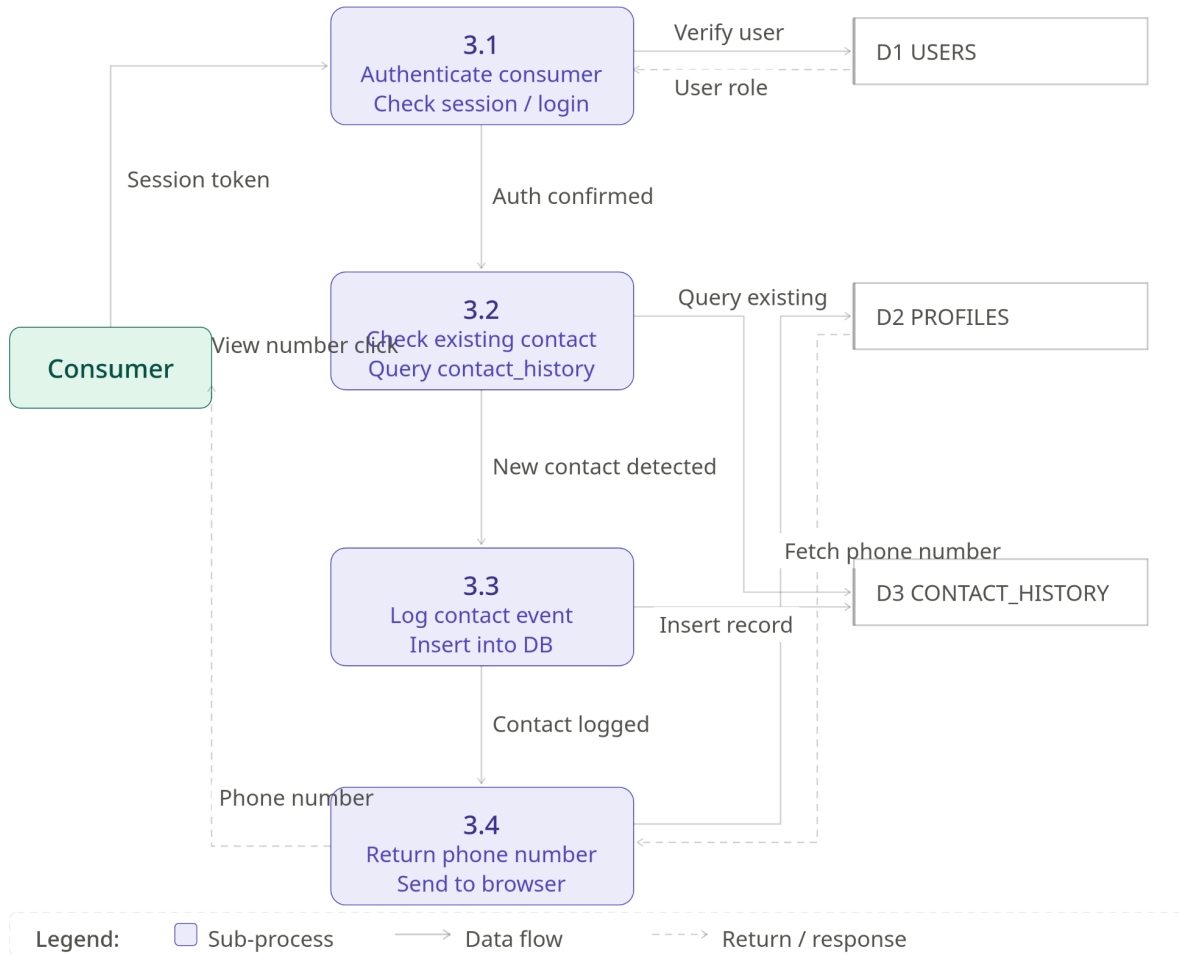
## 11.1 Level-0 DFD :



## 11.1 Level-1 DFD :

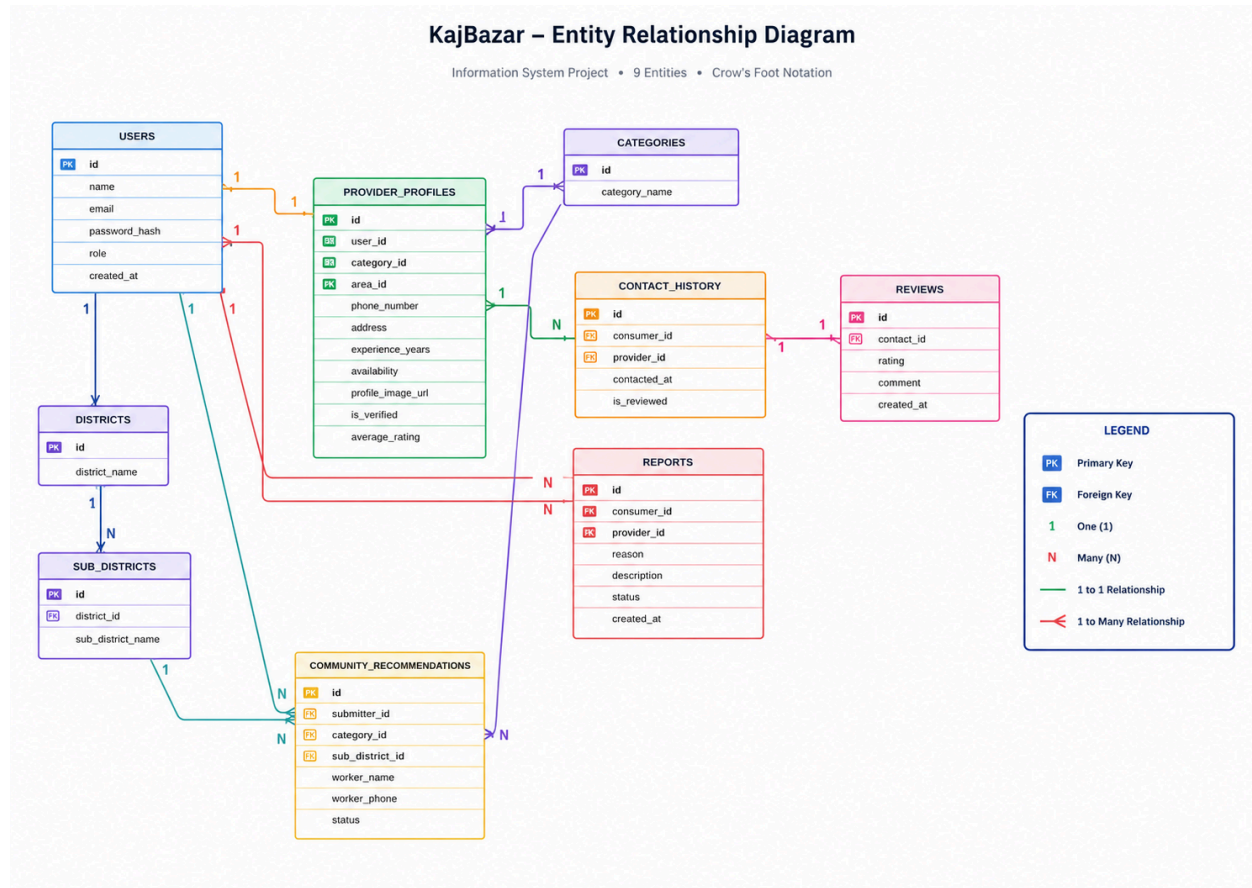
### Data flow diagram (DFD) — level 1

Process 3.0: Reveal phone number — KajBazar



## 12. Simple ER Diagram (Conceptual)

The conceptual ER diagram below illustrates the major entities of KajBazar and their relationships without including detailed attributes. It provides a high-level overview of the system's data structure and how the primary entities interact with one another.



## 13. Conclusion

KajBazar addresses the common challenge of connecting consumers with reliable local skilled workers by providing a centralized and community-driven service provider directory. Through features such as worker verification, location-based search, direct contact, ratings and reviews, and community recommendations, the platform aims to improve service accessibility while increasing visibility and employment opportunities for local service providers.

The proposed scope focuses on the core functionalities required for the initial release, making the project feasible to complete within a semester-long System Analysis and Design Sessional. Furthermore, the system is designed with scalability in mind, allowing future enhancements such as online service booking, integrated payment systems, real-time messaging, and mobile application support.